

M52: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

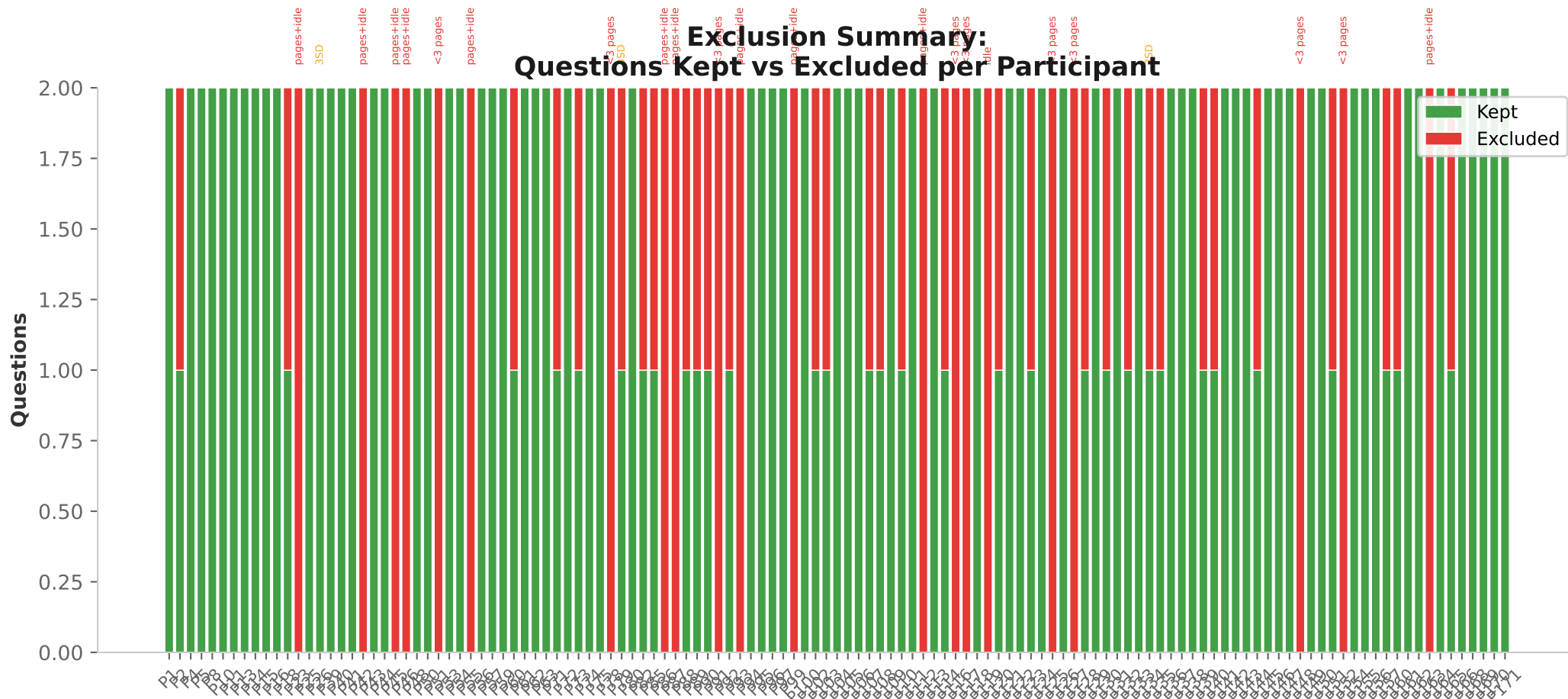
No standardization (z-score) applied

Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 65 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 27 excluded
3. Participants > 3 SD from mean on any DV: 3 excluded

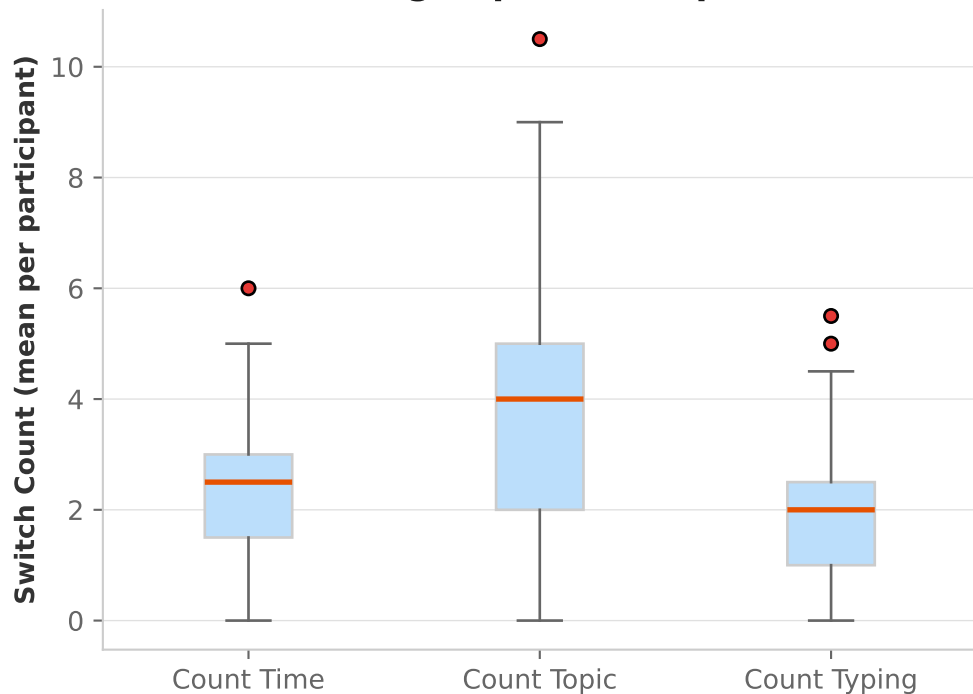
Result: 176 / 250 questions retained

Final sample: N = 101 participants

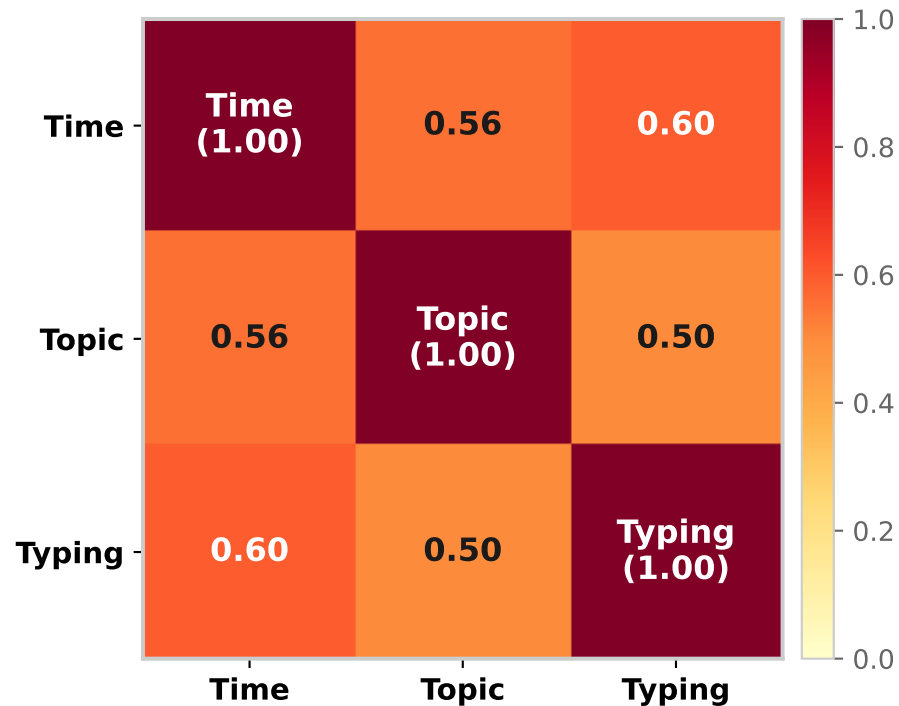


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

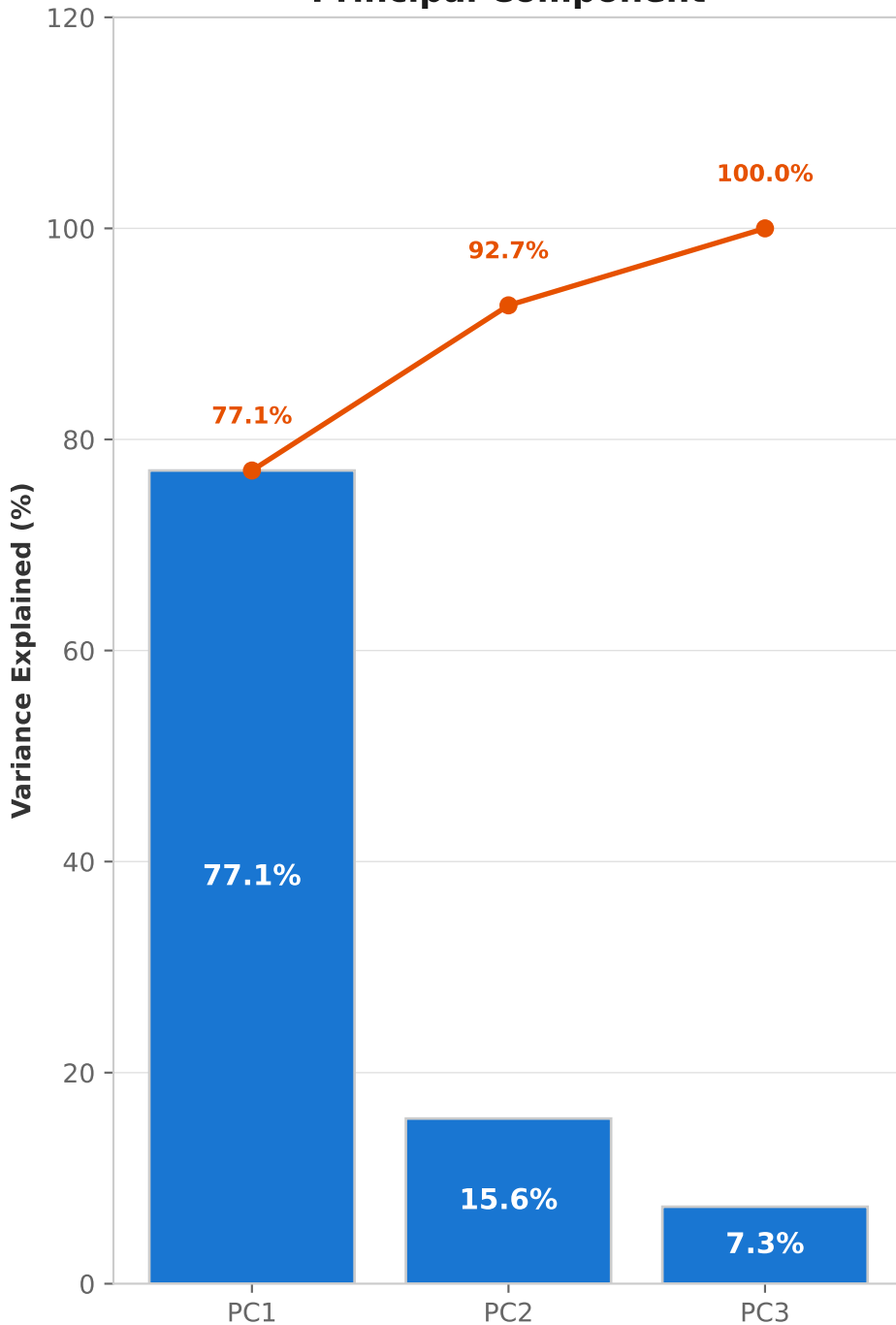


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (77.1% variance):

Count Topic	+0.879	+++++
Count Time	+0.374	+++++
Count Typing	+0.297	++++

PC2 (15.6% variance):

Count Time	+0.682	+++++
Count Typing	+0.554	+++++
Count Topic	-0.478	-----

PC3 (7.3% variance):

Count Typing	+0.778	+++++
Count Time	-0.629	-----
Count Topic	+0.004	

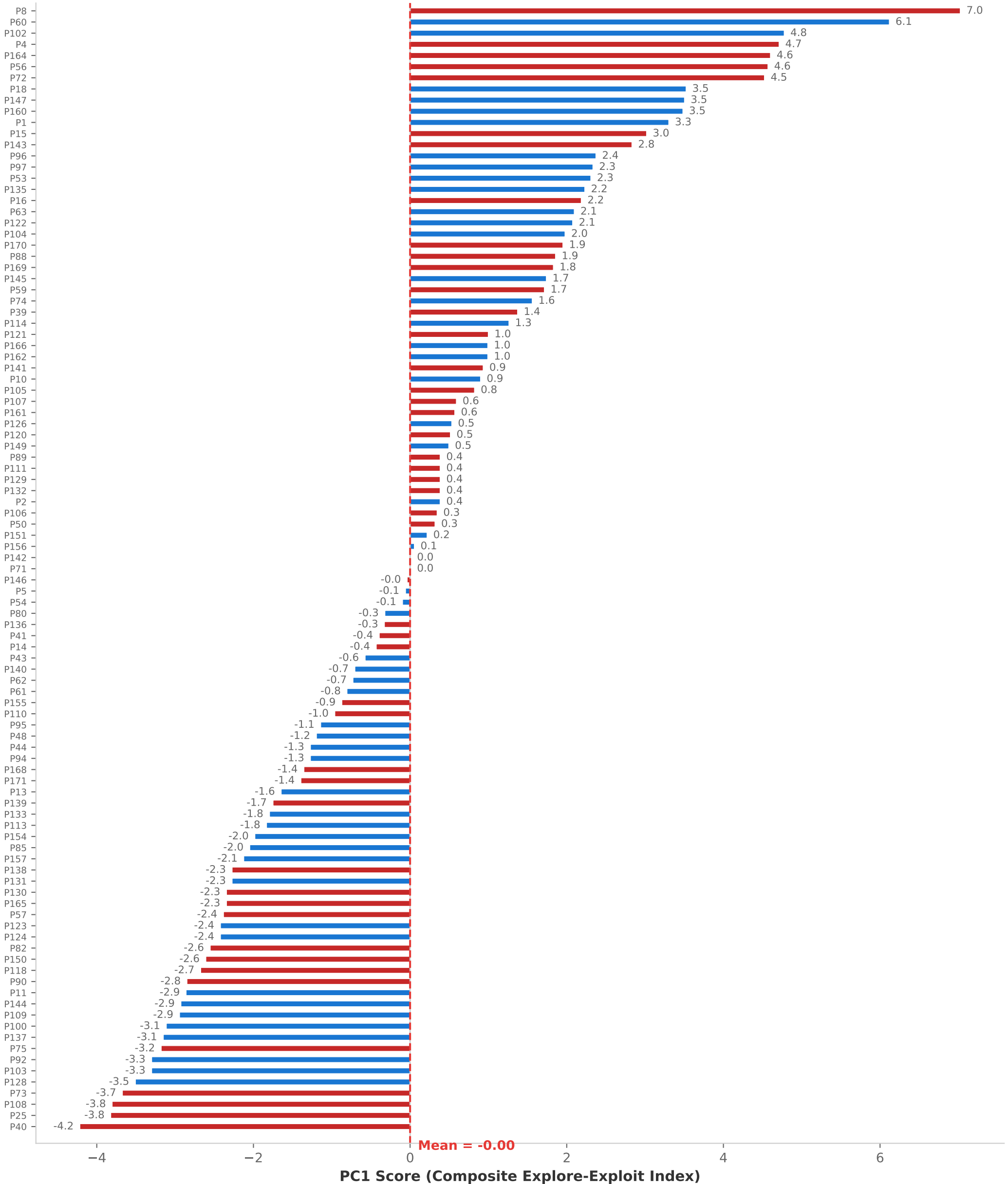
Interpretation:

PC1: Primarily reflects topic switching
(loading = +0.879)
Higher score = more topic transitions

PC2: Primarily reflects time switching
(loading = +0.682)
Higher score = more time transitions

PC3: Primarily reflects typing switching
(loading = +0.778)
Higher score = more typing transitions

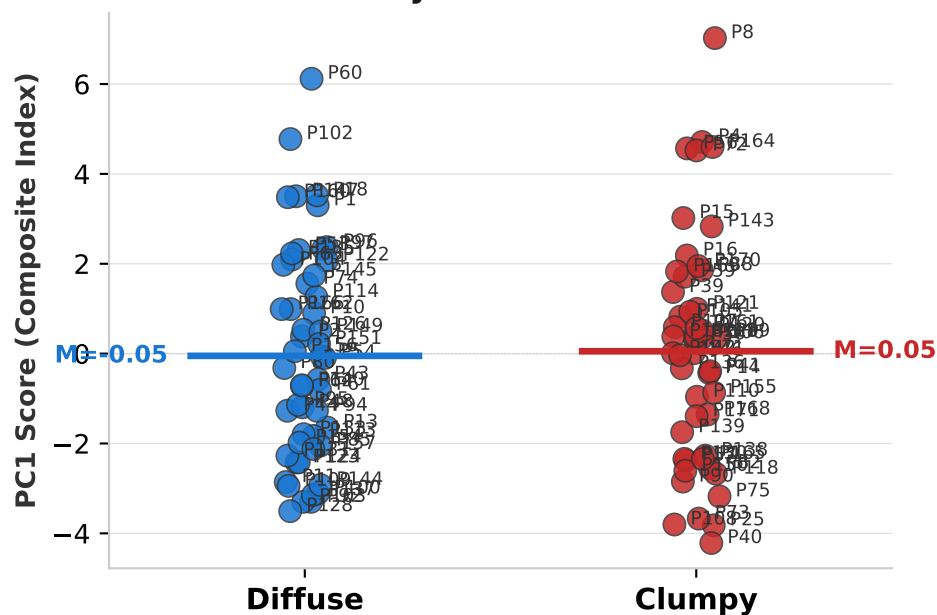
Individual PC1 Scores: Composite Explore-Exploit Index



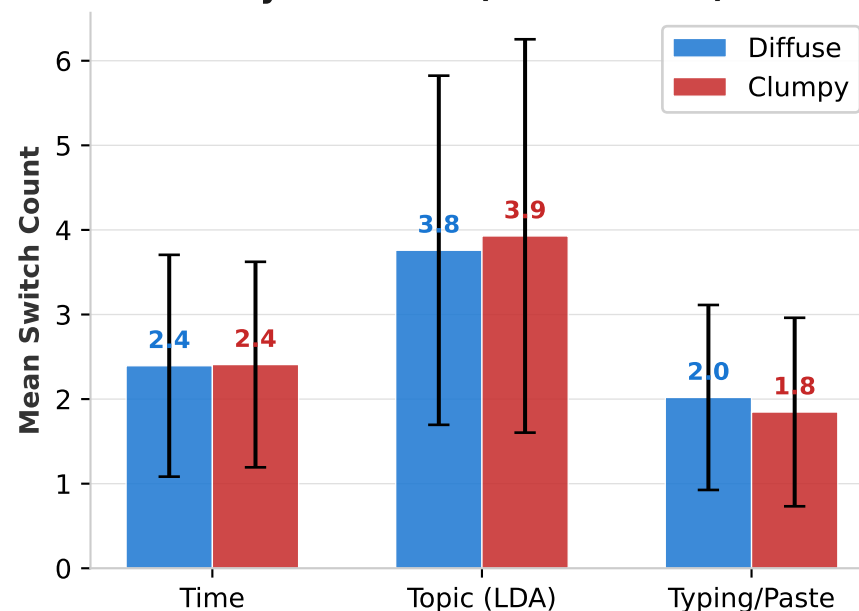
Condition Comparison: Clumpy vs Diffuse

N = 101 (Diffuse: 52, Clumpy: 49)

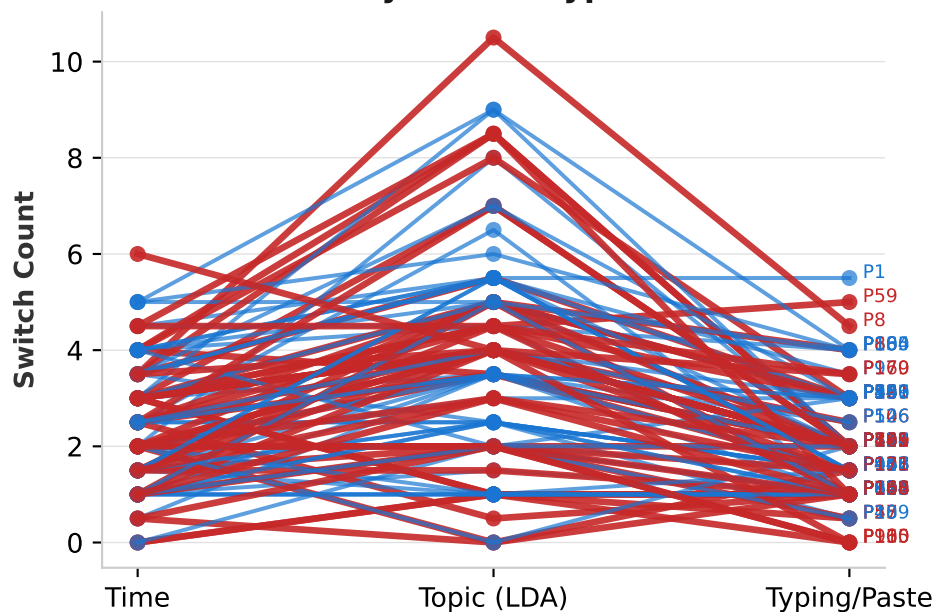
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean +/- SD)



Individual Profiles
by Switch Type



Conditions

- Diffuse (n=52)
- Clumpy (n=49)

Condition Comparison: Descriptive Statistics & t-Tests

N = 101 (Diffuse: 52, Clumpy: 49)

DESCRIPTIVE STATISTICS

Variable	Diffuse (n=52)	Clumpy (n=49)
Time	M = 2.39, SD = 1.31	M = 2.41, SD = 1.21
Topic (LDA)	M = 3.76, SD = 2.06	M = 3.93, SD = 2.33
Typing/Paste	M = 2.02, SD = 1.09	M = 1.85, SD = 1.11
PC1 (composite)	M = -0.05, SD = 2.34	M = 0.05, SD = 2.52

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	-0.06	99.0	0.9559	+0.01	n.s.
Topic (LDA)	-0.39	96.0	0.7008	+0.08	n.s.
Typing/Paste	0.78	98.4	0.4351	-0.16	n.s.
PC1 (composite)	-0.21	97.2	0.8329	+0.04	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +0.10
Clumpy condition shows more explore-exploit switches
Effect size (Cohen's d): +0.04
Result is NOT statistically significant (p = 0.8329)
Low statistical power due to small sample (N = 101)

Largest sub-variable effect: Typing/Paste
d = -0.16, p = 0.4351

Note: Typing metric uses answer_snapshot bursts + paste events.